
XCorr Trigger Timing

Relative to raw pulse rising edges

Manuel Arroyave

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Main Result

Timing decomposition

$$\Delta n(P) = n_{\text{trig}}(P) - n_{\text{edge}} = n_{\text{latency}} + n_{\text{walk}}(P).$$

- ▶ n_{edge} : raw rising-edge timestamp.
- ▶ $n_{\text{trig}}(P)$: xcorr trigger timestamp at PE threshold P .
- ▶ Fixed xcorr latency: 63 samples = 1008 ns.
- ▶ Residual walk: $n_{\text{walk}}(P)$, the threshold-dependent part.

Data Sources

Raw data

- ▶ Real waveform:
data/input/run039344_ch35.bin
- ▶ Samples: unsigned 14-bit ADC words in 16-bit little-endian storage
- ▶ Sampling clock: 62.5 MHz
- ▶ Sample period: 16 ns

Study location: ~/repo/daphne_mezz_xc_sim

Raw dynamic range

$$A_{\max} = 16\,383 - 2692 = 13\,691 \text{ ADC}$$

$$1 \text{ PE} = 10 \text{ ADC} \quad \Rightarrow \quad A_{\max} \approx 1369 \text{ PE}$$

Study Artifacts And PE Calibration

Code and outputs

- ▶ Timing script:
`scripts/plot_xcorr_delay_vs_pe.py`
- ▶ XCorr trace:
`data/output/analysis/run039344_ch35_rtl_thr2k_len1024_xcorr.txt`
- ▶ Summary CSV: `data/output/analysis/xcorr_threshold_timing/run039344_ch35_delay_vs_pe_summary.csv`

PE threshold conversion

The saturated-pulse plateau gives

$$1 \text{ PE}_{\text{thr}} = 581.9 \text{ xcorr counts.}$$

Plots

- ▶ Residual vs PE threshold
- ▶ Example pulses with edge marks

What Is Compared

Raw reference timestamp

For each pulse:

$$n_{\text{edge}} = \arg \max_{n < n_{\text{peak}}} \frac{d}{dn} (x[n] - \text{median}(x)).$$

This is the peak finder on the rising edge of the raw waveform.

XCorr trigger timestamp

For PE threshold setting P :

$$T(P) = P \cdot 581.9.$$

$$n_{\text{trig}}(P) = \min n$$

with $y[n] > T$, $y[n-1] > T$, and $y[n-2] \leq T$.
This mirrors the two-cycle threshold-crossing rule used by the emulator.

Important alignment point. The xcorr timestamp is produced by a causal matched-filter pipeline. It is not in the same phase reference as the raw ADC sample.

Causal Matched Filter

Filter form

$$y[n] = \sum_i h_i x[n - d_i], \quad d_i \simeq 3i + 4.$$

Where the delay comes from

- ▶ RTL-style transposed FIR tap i appears at approximately $3i + 4$ samples.
- ▶ The xcorr phase follows the template centroid, not the raw leading edge.

Template centroid

Using the negative template lobe:

$$\tau_h = \frac{\sum_{h_i < 0} (-h_i)(3i + 4)}{\sum_{h_i < 0} (-h_i)} = 63.9 \text{ samples.}$$

$$63 \text{ samples} \cdot 16 \text{ ns} = 1008 \text{ ns.}$$

Subtract The Fixed Latency

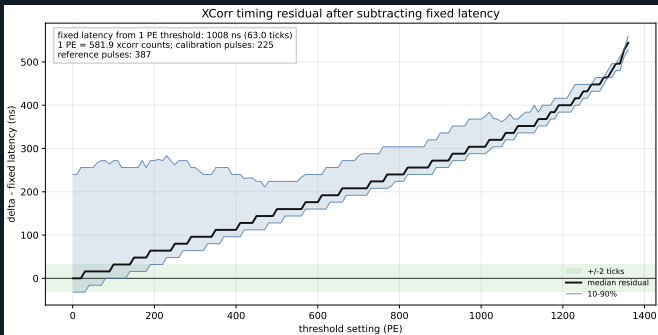
Latency-corrected residual

$$r(P) = [n_{\text{trig}}(P) - n_{\text{edge}}] - 63 \text{ samples.}$$

$$r_{\text{ns}}(P) = 16 \, r(P).$$

- ▶ The fixed latency is estimated from the 1 PE threshold median: 1008 ns= 63 sample.
- ▶ The residual is the threshold-dependent walk that matters for timing.
- ▶ The shaded green band in the next plot is ± 2 ticks = ± 32 ns.

Residual Timing Walk Versus PE Threshold



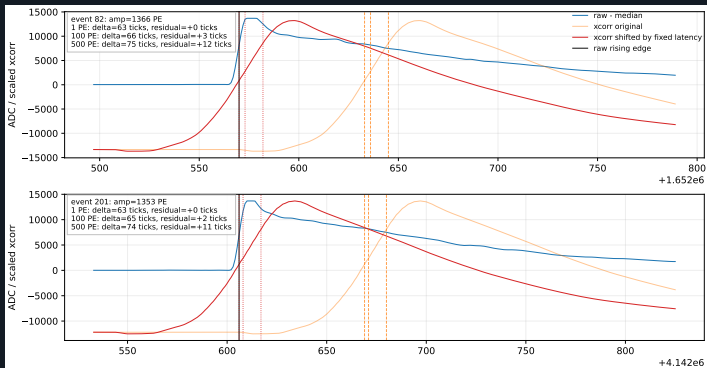
Black line: median residual. Blue band: 10–90% interval. Green band: ± 2 ticks.

Statistical Summary

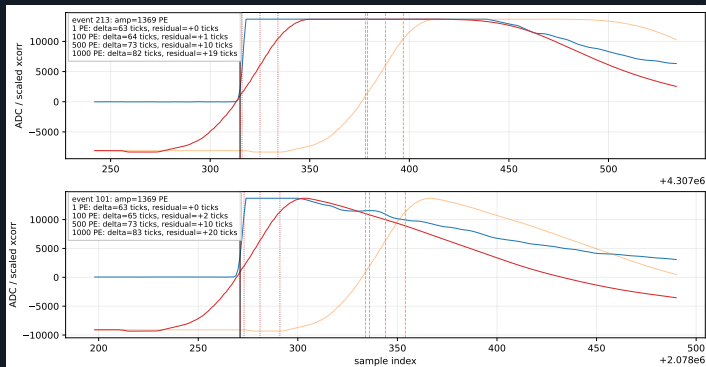
Threshold (PE)	Matched pulses	Residual (ticks)	Residual p10 (ticks)	Residual p90 (ticks)
1	386	0	-2	15
10	384	0	-2	15
20	382	0	-2	16
100	364	2	0	17
200	331	4	2	17
500	263	10	8	14
800	230	15	14	19
1000	199	19	18	23
1200	162	25	24	26
1300	138	29	28	29

Interpretation: low-threshold median residual is ≤ 2 ticks; large PE thresholds walk later because the trigger waits for a higher point on the xcorr lobe

Example Pulses: First Pair



Example Pulses: Second Pair



Why Threshold Walk Appears

Crossing equation. Near a pulse, after latency alignment:

$$y_{\text{aligned}}[k] \approx A g[k],$$

where A is pulse amplitude and $g[k]$ is the normalized xcorr lobe.

$$k_T = \min\{k : A g[k] > T\}.$$

- ▶ If T is small, k_T is close to the leading part of the xcorr lobe.
- ▶ If T is large, k_T moves up the lobe toward the peak.
- ▶ For saturated pulses, raising T from low PE to ~ 1000 PE moves the median crossing by about 19 ticks after latency correction.

Practical Consequence

Comparator behavior

The timing comparator must either subtract the fixed latency, use a constant-fraction-like timing reference, or constrain threshold settings to the region where residual walk is acceptable.

- ▶ The 63-sample term is a phase/latency offset.
- ▶ The residual walk is the physics/algorithm behavior that changes with threshold.
- ▶ The PE scan identifies the threshold boundaries where that residual leaves the acceptable timing band.

Reproducibility Checklist

Commands used for the PE scan

1. Run or reuse the xcorr trace from `daphne_mezz_xc_sim`.
2. Execute:

```
python3 scripts/plot_xcorr_delay_vs_pe.py
```

3. Use the generated summary CSV, residual plot, and pulse examples.

Validation anchors

- ▶ 62.5 MHz $\Rightarrow$ 16 ns per sample.
- ▶ Raw dynamic range: 13 691 ADC $\approx$ 1369 PE.
- ▶ Filter/template centroid: 63.9 samples.
- ▶ Measured fixed latency from data: 63 samples.

Generated Files

Analysis

- ▶ `data/output/analysis/xcorr_threshold_timing/run039344_ch35_delay_vs_pe_summary.csv`
- ▶ `data/output/analysis/xcorr_threshold_timing/run039344_ch35_pe_calibration.csv`

Figures

- ▶ `data/output/plots/xcorr_residual_vs_pe_run039344_ch35.pdf`
- ▶ `data/output/plots/xcorr_example_pulses_run039344_ch35.pdf`

Bottom Line

1. The measured xcorr time is fixed matched-filter latency plus residual threshold walk.
2. The fixed latency is ≈ 63 samples, confirmed mathematically and from the 1 PE data point.
3. The timing quantity of interest is the residual:

$$n_{\text{trig}} - n_{\text{edge}} - 63.$$

4. Low thresholds have near-zero median residual.
5. High thresholds introduce deterministic threshold walk because the crossing point moves later on the xcorr lobe.